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14 February 1956
SUPERSEDING
MIL–W–13725 (ORD)
22 October 1954

MILITARY SPECIFICATION

WATCH, POCKET (7 SECONDS DAILY RATE)

This specification was approved by the Departments of the Army, the Navy, and the Air Force, and is mandatory for use of procurement services of the respective Departments

1. SCOPE

1.1 This specification covers a pocket watch used on military missions, such as railroad operations, timing celestial observations and for comparing purposes where accurate time-keeping is essential.

1.2 Class. Pocket watches shall be of the following classes as specified in the contract (see 6.1):

Class 1—Stem wind, lever set.

Class 2—Stem wind, stem set, second setting mechanism.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATION

MILITARY

MIL–F–13926 — Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL–P–11446 — Packaging of Watches, Pocket, Stop and Wrist.

MIL–P–17207 — Preservation and Packaging of Horological Repair Parts and Accessories Associated With Navigational Timepieces.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MS35450 — Bearing, Jewel; Bar Hole.

MS35451 — Bearing, Jewel, Center Bar Hole.

MS35452 — Bearing, Jewel; Olive Hole.

MS35453 — Bearing, Jewel; Olive Hole (cone cut).

MS35454 — Bearing, Jewel; Endstone.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FED. SUP. CLASS
6645

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3. REQUIREMENTS

3.1 Material. Where a definite material is not specified it shall be of a composition and quality that is normally used in the manufacture of this specific quality of watch. All materials shall be of uniform quality and free of any defects which might impair the functioning of the watch.

3.2 Movement.

3.2.1 *General.* The movement shall be of standard commercial manufacture and assembly, size 16, ¾ plate or bridge of rigid nickel construction, double roller escapement, Breque hairspring, micrometric regulator, and the complete movement shall fit into the case snugly without interference. The plates and bridges shall be free of burrs, sharp edges, rough surfaces and defects. All steel parts and screws not protected by lubricant shall be black oxide finished, nickel plated or rust proofed in some approved manner except those parts, the proper functioning of which, would be detrimentally affected.

3.2.2 *Mainspring.* The mainspring shall be of stainless alloy material and shall drive the completed movement a minimum of 36 hours without rewinding. The spring shall be of type and quality used in best commercial watches.

3.2.3 *Hairspring and balance wheel unit.* The movement shall have a temperature compensating hairspring and a nonmagnetic, solid, monometallic balance wheel. Both shall be made of materials such that the combination will not be permanently affected, except by hysteresis, in the presence of a magnetic field.

3.2.4 *Jewels.* The movement shall have not less than 17 jewel bearings, located at bearing points most essential to reduce friction of train and escapement parts. Jewels shall be secured in the plate or bridge by means of friction fit or screws. Endstone jewels shall be secured in such a manner to facilitate cleaning and reoiling. Bar hole, center bar hole, olive hole, olive hole (cone cut) and endstone jewel bearings used in watch movement shall be in accordance with the applicable Military Standards listed under 2.1.

3.2.5 *Winding and setting.*

3.2.5.1 *Class 1.*

3.2.5.1.1 *Winding stem.* The winding stem shall be carefully adjusted and alined with the movement and a winding crown shall fit snugly on the pendant of the case. Entrance for the stem shall be located at a point above the numeral 12 on the dial and construction shall insure smooth operation. Continuous winding operations shall not adversely affect the timekeeping properties of the watch.

3.2.5.1.2 *Setting lever.* The setting lever shall be separate from the winding stem and located inside of the case assembly. Continuous setting operations shall not adversely affect the timekeeping properties of the watch.

3.2.5.2 *Class 2.*

3.2.5.2.1 *Winding and setting.* The movement shall be stem wound and stem set, with the winding and setting mechanism incorporated as part of the movement and operating independently of the case. Entrance for the stems shall be located at a point above the numeral 12 on the dial. The stem and crown shall be of dustproof and moisture-proof construction and shall insure smooth operation during winding and setting. Winding and setting operations shall not adversely affect the time-keeping properties of the watch.

3.2.5.2.2 *Second setting mechanism.* The second setting mechanism incorporated in the movement shall be operated through

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the stem and crown of the watch in such a manner that a flexible spring will come in contact with the rim of the balance when stem is pulled into setting position and be immediately released when stem is returned to winding position. The second setting mechanism shall be so constructed that it will not impair time-keeping properties of the watch by repeated contacts with the balance wheel.

3.2.6 *Regulator.* The regulator shall be at the center of adjustment within plus or minus 20 pecent of its total range.

3.3 **Case assembly.** The case shall be of nickel or an approved nickel alloy, round, with antique bow, threaded back and bezel (the class 2 watch bezel may be integrally cast and machined with case), without sharp edges. The bezel shall be fitted with a non-breakable, permanently clear, nonfading, transparent, plastic crystal and shall not be adversely affected as a result of exposure to extreme temperature tests. Assembly of the crystal to the bezel shall be dustproof and moistureproof. The outside surface of the case shall have a bright nickel finish.

3.4 **Dial.** The face of the dial shall be of white porcelain, single or double sunk, of the Montgomery safety numeral type and the manufacturer's name shall show on the face of the dial in an inconspicuous manner. All markings shall be clear, contrasting and legible. All numerals shall be Arabic, vertical with the stem; hours, minutes, seconds and 10 second periods shall be plainly and distinctively marked in black. Numerals and marks shall be flush with the surface of the dial or slightly depressed.

3.4.1 *Class 2 watch seconds dial.* The seconds dial orbit in the class 2 watch (1.2) shall be at least 0.580 inch in diameter and shall be divided into seconds with 10 second divisions indicated in Gothic numerals. Figures and marking shall be an integral part of the basic porcelain and shall be black, with figures flush with the surface of the porcelain or slightly depressed.

3.5 **Hands.** The hour, minute and second hands shall be of blued steel. The hour hand shall be of heavy spade type. The minute hand shall be distinctly different in type and length from the hour hand. The second hand shall be a plain tapered type of sufficient length to provide accurate readings with the second markings on the dial.

3.5.1 In the finished watch, the hands shall be essentially parallel to the dial and to each other with sufficient clearance in each instance to prevent rubbing or binding. They shall rotate concentrically with respect to the applicable minute and second dial tracks.

3.6 **Magnetism.** While running, the watch shall not be adversely affected when subjected to a 10 Gauss magnetic field for a period of at least 5 seconds and shall subsequently pass the accuracy requirements of 3.7.1 and 3.7.2.1.

3.7 **Accuracy.** The following tolerances shall not be exceeded when determined by compliance with 3.8.

3.7.1 *Mean daily rate.* The mean daily rate for period 1 shall not exceed 7 seconds.

3.7.2 *Difference in mean daily rates.*

3.7.2.1 *Periods 1 and 2.* The difference in the mean daily rate for periods 1 and 2 shall not exceed 12 seconds.

3.7.2.2 *Periods 2 and 3.* The difference in the mean daily rate for periods 2 and 3 shall not exceed 5 seconds.

3.7.2.3 *Isochronism.* After running 24 hours to allow settling out and rewinding, the difference in period 2, between the amount gained or lost in the first 24 hours, and twice the amount gained or lost in the first 12 hours, shall not exceed 3 seconds.

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3.7.2.4 *Periods 2, 4 and 5.* The difference in the mean daily rates of periods 2, 4 and 5 shall not exceed 8 seconds.

3.7.3 *Recovery rate.* The difference between the mean daily rates for periods 1 and 6 shall not exceed 4 seconds.

3.8 **Temperature and position.** The watch shall be capable of meeting the accuracy requirements of 3.7 for the following temperatures and positions as indicated by periods and number of days.

Period	*Days*	*Temperature*	*Position*
1	3	75°F.±3°F.	Pendant up
2	3	75°F.±3°F.	Dial up
3	3	75°F.±3°F.	Dial down
4	3	40°F.±2°F.	Dial up
5	3	95°F.±2°F.	Dial up
6	3	75°F.±3°F.	Pendant up

3.9 **Interchangeability.** Like parts shall be interchangeable in all like watches supplied by one manufacturer and shall not adversely affect timekeeping qualities except for minor adjustments.

3.10 **Identification.** Unless otherwise specified, each watch shall legibly show on the movement (barrel bridge or train bridge, or both) the manufacturer's name and serial number and manufacturer's model or grade number. Each case shall be distinctly marked on the exterior of the back with, "GRADE No. I," below the words "U. S. GOVERNMENT."

3.11 **Workmanship.** Standards of workmanship shall be such that the watches will meet the requirements of 3.1 through 3.10. All components, subassemblies, parts, and the final assembly shall meet the applicable workmanship requirements specified in MIL–F–13926.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Pilot lot.** A pilot lot representing initial production shall be submitted for complete test and acceptance in accordance with all requirements herein. The pilot lot size shall be determined by the Government inspector and the watches shall be representative of design, material and workmanship expected in normal production applying best commercial practices.

4.2 **Sampling.** Sampling, including lot size shall be in accordance with MIL–STD–105 and the acceptable quality levels specified herein.

4.3 **Inspection.** The Government inspector shall ascertain prior to and during assembly as well as after final assembly that production components, subassemblies and assemblies are in accordance with the test procedures of 4.4 and the acceptable quality levels and classification of defects outlined herein to verify conformance with the requirements of this specification.

4.3.1 *Acceptable quality levels (AQL's).* Inspection tests for acceptance shall be conducted in accordance with the following classification of defects and acceptable quality levels expressed in percent defective.

Major *AQL 2.5*		*Test procedure*	*Method of inspection*
1. Case assembly	(3.3)	4.4.10	Visual
2. Dial	(3.4)	4.4.11	Visual
3. Hands	(3.5)	4.4.7 & 4.4.11	Visual & Functional
4. Mean daily rate	(3.7.1)	4.4.13	Functional
5. Periods 1 & 2	(3.7.2.1)	4.4.13	Functional
6. Periods 2 & 3	(3.7.2.2)	4.4.13	Functional
7. Mainspring	(3.2.2)	4.4.3 & 4.4.13	Functional

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Major	AQL 2.5 inspection level L8 of table VII		Test procedure	Method of inspection
1.	Periods 2, 4 & 5	(3.7.2.4)	4.4.13	Functional
2.	Recovery rate	(3.7.3)	4.4.13	Functional
3.	Isochronism	(3.7.2.3)	4.4.13	Functional
4.	Regulator	(3.2.6)	4.4.9	Visual

Minor	AQL 4.0		Test procedure	Method of inspection
41.	Crystal	(3.3)	4.4.10.1	Visual
42.	Identification	(3.4 & 3.10)	4.4.15	Visual
43.	Workmanship	(3.2.1 & 3.11)	4.4.16	Visual

4.4 Test procedures.

4.4.1 *Material.* Where defects or inferior quality is evident in component parts and the Government inspector deems a material analysis necessary to determine compliance with 3.1, the contractor will be requested to submit samples or specimens to the contracting officer for analysis and approval. Where a technical analysis is desired, the samples or specimens should be forwarded to the Commanding General, Frankford Arsenal, Attention of Industrial Division, Fire Control Instrument Group, Philadelphia 37, Pa.

4.4.2 *Movement.* Analysis and inspection of the movement shall be made prior to and at final assembly to determine compliance with 3.2.1.

4.4.3 *Mainspring.* Certification as to material and quality shall be obtained in addition to the performance test of 4.4.13 to insure the minimum running time without rewinding for compliance with 3.2.2.

4.4.4 *Hairspring and balance wheel unit.* The hairspring and balance wheel unit requirement of 3.2.3 shall be considered acceptable if the watch is capable of meeting the requirements of 3.7 and 3.8.

4.4.5 *Jewels.* One percent of the watches under contract, but not less than three watches shall be inspected for compliance with 3.2.4.

4.4.6 *Winding stem test.* The 24 hour daily winding during accuracy tests of 4.4.13 shall be utilized to determine compliance with 3.2.5.1.1 or 3.2.5.2.1.

4.4.7 *Stem or lever setting test.* Six settings shall be made in two hour increments to insure compliance with 3.2.5.1.2 or 3.2.5.2.1.

4.4.8 *Second setting mechanism.* At least 5 settings of the second setting mechanism shall be made to insure compliance with 3.2.5.2.2.

4.4.9 *Regulator.* The regulator adjustment shall be inspected after completion of all functional tests to insure compliance with 3.2.6.

4.4.10 *Case assembly.* All component parts and the completed case assembly shall be visually inspected for compliance with 3.3 and 3.11.

4.4.10.1 *Crystal.* Upon completion of accuracy tests for periods 4 and 5 (3.8), the crystal shall be inspected to insure compliance with 3.3.

4.4.11 *Dial and hands.* A careful visual and physical examination shall be made to determine compliance with 3.4.3.5 and 3.12.

4.4.12 *Magnetism test.* One percent of the watches under contract but not less than three watches shall meet the requirements of 3.6.

4.4.13 *Accuracy tests.* Prior to the accuracy tests the watch shall be conditioned by being fully wound and shall have run a minimum of 36 hours without rewinding to determine compliance with 3.2.2. To determine compliance with 3.7 and 3.8, the watch shall be wound at the beginning of each test and each

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24 hours thereafter for the duration of the tests. The mean daily rate shall be recorded for each period stipulated in 3.7.

4.4.13.1 *Temperature and position tests.* When conducting accuracy tests, in accordance with 3.7, the watches will be permitted to be conditioned for 24 hours prior to conducting tests after each temperature change.

4.4.14 *Interchangeability tests.* The contracting officer should schedule interchangeability tests when it is deemed necessary to insure compliance with 3.9.

4.4.15 *Identification.* A visual inspection shall be made to determine compliance with 3.10.

4.4.16 *Workmanship.* Workmanship shall be inspected at the discretion of the Government inspector during in-process and on the end item to insure that watches produced are in accordance with 3.11.

4.4.17 *Resubmission after rejection.* Watches rejected for failure during or after testing may be reworked and resubmitted for inspection by furnishing data concerning corrective action taken and measures employed to prevent recurrence of the same deficiencies. Watches rejected on a retest shall not be resubmitted without specific approval of the contracting officer.

5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.** Preservation, packaging and packing for domestic and overseas shipment shall be in accordance with MIL–P–11446 and applicable protection Levels (A, B or C).

5.2 **Repair parts and accessories.** Repair parts and accessories shall be preserved and packaged in accordance with MIL–P–17207.

5.3 **Marking.** Marking shall be in accordance with MIL–ST–129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Class required (1.2).

(c) Packaging level (A, B or C) and Packing level (A, B or C).

(d) Repair parts or accessories as well as complete watches (5.2).

(e) Special identification (3.10).

(f) Interchangeability tests (4.4.14).

6.2 **Grade.** Inasmuch as military watches of 10 seconds daily rate accuracy or better are designated Grade I, this pocket watch is accordingly identified as Grade I (3.10).

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fast that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Ordnance Corps
Navy—Bureau of Ships

Other Interest:
Army—TQE

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